Aim: To install, explore, and test simulation tools such as Tinkercad, CupCarbon, Contiki, and iFogSim (iKooja) for designing and analyzing Micromouse and IoT-based systems.

1. Software Installation & Setup (Tinkercad)

Objectives:

To install and verify simulation environments used for embedded systems and IoT projects.

Tools Used:

- Arduino IDE
- TinkerCad
- USB Drivers for Arduino Nano

Steps Followed:

1. Open a web browser (Google Chrome recommended).
2. Navigate to the official Tinkercad website.
3. Create a new account or log in using existing credentials.
4. Select “Circuits” from the dashboard.
5. Click on “Create New Circuit”.
6. Verify the environment by adding an Arduino Uno and running a basic simulation.

2. Project Design

Objective: To design and simulate Micromouse control logic using virtual components.

Steps Followed:

- Open Tinkercad Circuits.
- Click on “Create New Circuit”.
- Add the following components:
    - Arduino Uno
    - DC Motors (representing robot wheels)
    - Motor driver (simulated using basic components)
    - Input switches (to simulate sensors)
- Connect components:
    - Motors connected to digital/PWM pins
    - Switches connected to input pins
    - Proper power and ground connections established
- Arrange components neatly for clarity.
- Open the code editor and write Arduino logic.

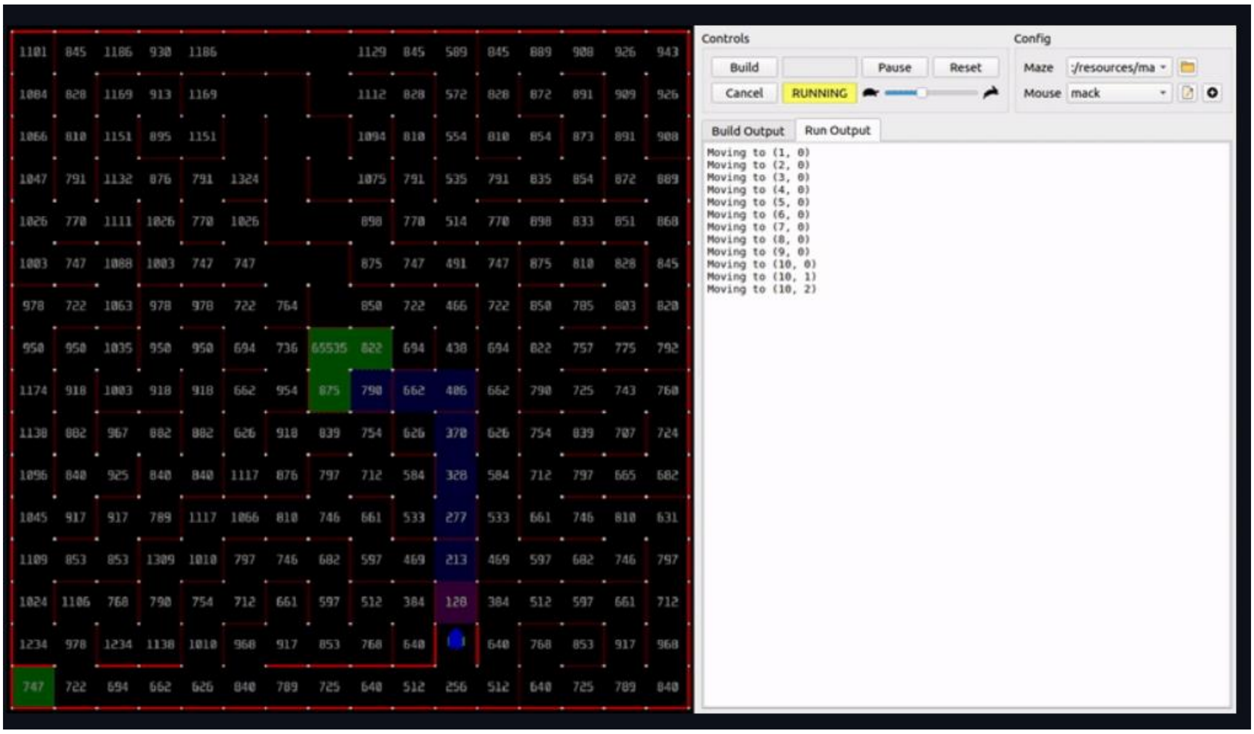
Components Used:

Component	Specification	Application

Arduino Nano	ATmega328P	Main controller
TB6612FNG Motor Driver	Dual H-Bridge	Motor control
N20 Gear Motors (1000 RPM)	DC Motors	Robot movement
TCRT5000 IR Sensors	Reflective IR	Wall detection
18650 Li-ion Battery	3.7V each	Power supply
LM2596 Buck Converter	Voltage regulator	Stable 5V output
Buzzer	Active buzzer	Alert indication
LED	Indicator	Status display
Push Button	Input switch	Start control

3. Simulation / Testing

Objective: To validate logic and behavior before full hardware execution.



Steps Followed:

- Click on “Start Simulation” in Tinkercad.
- Provide input using switches (simulate walls).
- Observe system response:
  - No obstacle → motors move forward
  - Left blocked → robot turns right
  - Right blocked → robot turns left

   - All blocked → robot turns back
- Stop simulation and modify code if required.

Observation:

- Sensors correctly detected walls.
- Motors responded accurately to direction commands.
- Logic execution matched expected behavior.

4. Concept of the Project

The Micromouse is an autonomous robot designed to solve a maze efficiently using intelligent pathfinding.

Working Principle:

- Sensors detect walls in front, left, and right.
- Maze structure is updated dynamically.
- Flood Fill Algorithm calculates shortest path.
- Robot moves toward cell with minimum distance value.
- Process repeats until goal is reached.

Algorithm Used: Flood Fill Algorithm (BFS based)

Advantages:

- Autonomous navigation
- Efficient shortest path finding
- Real-time decision making
- Low-cost embedded system

5. Code Implementation

- Tools Used:
- Arduino IDE
- Embedded C/C++

Key Logic:

- Read sensor inputs
- Update maze map
- Apply flood fill algorithm
- Control motors based on direction

Code:

```
#define FL 4
#define FR 5
#define L  6
#define R  7
```

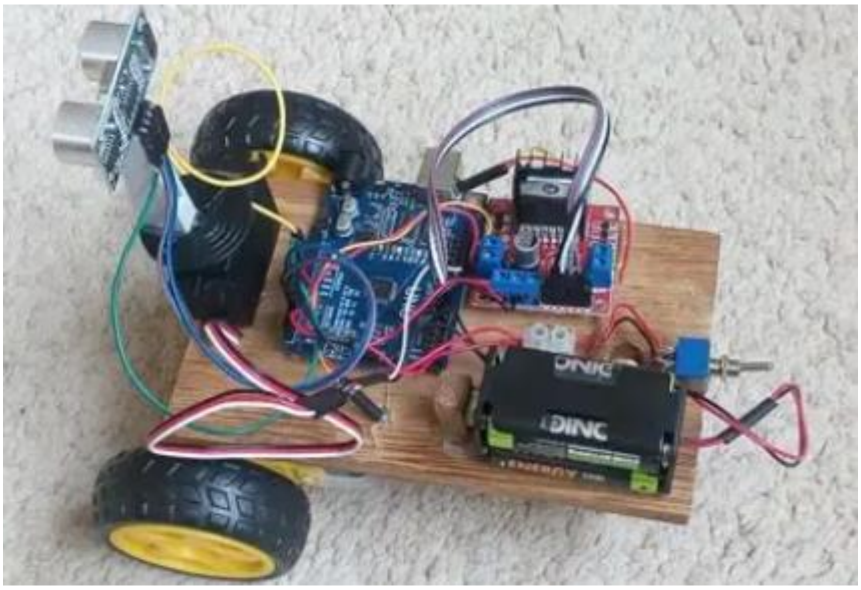
```
#define AIN1 2
#define AIN2 3
#define PWMA 9
#define BIN1 11
#define BIN2 12
```

```
#define PWMB 10
#define STBY 8

#define SPEED 170
#define CELL_TIME 500
#define TURN_TIME 300

bool wallF, wallL, wallR;
int dir = 0; // 0=N,1=E,2=S,3=W

void setup() {
  pinMode(FL, INPUT); pinMode(FR, INPUT);
  pinMode(L, INPUT);  pinMode(R, INPUT);

  pinMode(AIN1, OUTPUT); pinMode(AIN2,
OUTPUT);
  pinMode(BIN1, OUTPUT); pinMode(BIN2,
OUTPUT);
  pinMode(PWMA, OUTPUT); pinMode(PWMB,
OUTPUT);
  pinMode(STBY, OUTPUT);

  digitalWrite(STBY, HIGH);
}

void loop() {
  readSensors();
  decideMove();
}

void readSensors() {
  wallF = (digitalRead(FL)==LOW &&
digitalRead(FR)==LOW);
  wallL = (digitalRead(L)==LOW);
  wallR = (digitalRead(R)==LOW);
}

void decideMove() {
  if(!wallF) {
    moveForward();
  }
  else if(!wallL) {
    turnLeft();
    moveForward();
  }
  else if(!wallR) {
    turnRight();
    moveForward();
  }
  else {
    turnRight();
    turnRight();
  }
}

void moveForward() {
  digitalWrite(AIN1,HIGH);
digitalWrite(AIN2,LOW);
  digitalWrite(BIN1,HIGH);
digitalWrite(BIN2,LOW);

  analogWrite(PWMA,SPEED);
  analogWrite(PWMB,SPEED);

  delay(CELL_TIME);
  stopMotors();
}

void turnLeft() {
  digitalWrite(AIN1,LOW);
digitalWrite(AIN2,HIGH);
  digitalWrite(BIN1,HIGH);
digitalWrite(BIN2,LOW);

  analogWrite(PWMA,SPEED);
  analogWrite(PWMB,SPEED);

  delay(TURN_TIME);
  dir = (dir+3)%4;
  stopMotors();
}

void turnRight() {
  digitalWrite(AIN1,HIGH);
digitalWrite(AIN2,LOW);
  digitalWrite(BIN1,LOW);
digitalWrite(BIN2,HIGH);

  analogWrite(PWMA,SPEED);
  analogWrite(PWMB,SPEED);

  delay(TURN_TIME);
  dir = (dir+1)%4;
  stopMotors();
}

void stopMotors() {
  analogWrite(PWMA,0);
  analogWrite(PWMB,0);
}
```

6. Hardware Implementation

- Mounted motors and wheels on chassis base.
- Fixed caster wheel for balance.

- Installed IR sensors at proper angles.
- Connected motor driver to Arduino Nano.
- Built power circuit using LM2596.
- Verified 5V output using multimeter.
- Uploaded final code to Arduino.



Power Configuration:

- 2× 18650 batteries in series
- LM2596 regulates voltage to 5V
- Motor driver powered directly

Conclusion:

The Micromouse project demonstrates the integration of embedded systems, sensors, and algorithms to achieve intelligent autonomous navigation. The system is efficient, scalable, and suitable for robotics and IoT applications.